

The empirical recurrence for OEIS A234755: recovering a conjecture that is not written down

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Abstract

OEIS A234755 records “Empirical recurrence of order 55”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 65$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 55, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 55 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234755 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with 2×2 edge jumps all no more than +1 in one of the clockwise or counterclockwise directions or both.”. It has offset 1 and begins

8968, 646036, 41065150, 2683721676, 174451064914, 11352560005654, 738613242438722, 48057390396250352, ...

The entry states:

Empirical recurrence of order 55 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:24:03, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 257$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 65$ of the 257, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 65$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 514$ exact terms of the model returns order 55 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{55} c_i a(n-i),$$

c_1	65	c_{15}	−183222442962229	c_{29}	4205657292189457225	c_{43}	3499318703999176
c_2	386	c_{16}	294552218345283	c_{30}	−8159219193521117571	c_{44}	51917615645303512
c_3	−25139	c_{17}	2050061126827130	c_{31}	−4891888038438891340	c_{45}	−2695360658675880
c_4	−38422	c_{18}	−3426049329714142	c_{32}	9940890239977287928	c_{46}	−6682617778454304
c_5	3818001	c_{19}	−16434318376679997	c_{33}	4121807383451442606	c_{47}	565604175948284
c_6	93193	c_{20}	28211458281468692	c_{34}	−8950116238196313741	c_{48}	560997051636704
c_7	−312566085	c_{21}	94882472723248209	c_{35}	−2485955866445450724	c_{49}	−63084383258696
c_8	221511982	c_{22}	−166378579177782682	c_{36}	5961354274232566149	c_{50}	−27840010565800
c_9	15688276425	c_{23}	−395901548798838089	c_{37}	1043440905558834928	c_{51}	3801013822112
c_{10}	−17528906563	c_{24}	708136685488705218	c_{38}	−2932543496126557499	c_{52}	667326413344
c_{11}	−516822461274	c_{25}	1197230873519469786	c_{39}	−283537442452216101	c_{53}	−105511277824
c_{12}	703307979184	c_{26}	−2187931503961450400	c_{40}	1059455445418646567	c_{54}	−4333996544
c_{13}	11631225938657	c_{27}	−2630401249981243939	c_{41}	37911875683307314	c_{55}	804298752
c_{14}	−17581775871456	c_{28}	4933462485710191871	c_{42}	−277941754444689735		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 55, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $55 + 4$ terms removed returns order 55 as well, so $\deg q_{\min} = 55 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $55 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 55 that this sequence satisfies — and that family turns out to have exactly one member.

References

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